KARPAGAM ACADEMY OF HIGHER EDUCATION (DEEMED TO BE UNIVERSITY)

HACKSPRINT 2.0 CASE STUDY

AI-Powered Contractor Job & Work Order Management Platform

Team Name:	Sprint X Coders
Domain:	Smart Workforce & Job Management
Category:	Software

Strategic Problem Analysis

In the rapidly changing domain of property maintenance and workforce logistics, traditional manual processes create significant operational bottlenecks. The current operational landscape suffers from several foundational limitations that restrict growth and transparency:

- **Lack of Unified Workflow:** No centralized system exists to manage job lifecycles smoothly from initial upload to final invoicing.
- **Manual Job Discovery:** Contractors lack automated tools to efficiently filter and discover relevant job postings matching their specific expertise.
- **Fragmented Documentation:** Critical paperwork including work orders, design plans, and invoices are handled via disconnected tools.
- **Real-time Tracking Deficit:** Operational agents are unable to monitor the real-time progress of contractors once a job is actively assigned.
- **Inconsistent Data Entry:** A lack of standardized forms leads to non-uniform, error-prone, or incomplete job details.
- **Communication & Alerts:** The complete absence of real-time notifications forces users to manually refresh systems for updates.

User Pain Points

For Agents: Faced with a heavy manual review burden during contractor assignment, leading to delays. Communication gaps make it difficult to verify if a work plan has been initiated or completed.

Core Operational Challenges

- **"Black Box" Operations:** Complete lack of visibility into real-time field activities and location verification.

For Contractors: Experience inaccessible opportunities due to the lack of dedicated job discovery tools, along with invoicing friction that delays payments.

- **Fake Progress Claims:** Vulnerability to fraudulent reporting and payments for unperformed tasks.
- **Delayed Project Lifecycles:** Poor coordination consistently pushing timelines past deadlines.

The Next-Gen Solution

> **Project Vision:** Next-Gen AI Ecosystem for Property Maintenance & Workforce Logistics. A centralized digital ecosystem designed to seamlessly orchestrate the end-to-end lifecycle of property maintenance, transforming reactive operations into a predictive model.

The platform introduces four foundational pillars of innovation:

- **End-to-End Automation:** Consolidates job discovery, intelligent onboarding, and automated digital invoicing into a single, cohesive module.
- **AI-Driven Optimization:** Utilizes advanced matching algorithms to instantly pair the optimal contractor with a job based on skills, location data, and historical performance metrics.
- **Verifiable Transparency:** Introduces a "Live Progress Proof" feature with time-and-location-tagged media to completely eliminate fraudulent updates and build absolute trust.
- **Predictive Operations:** Forwards predictive models to analyze potential project delays and budget overruns before they occur.

Job Posting → AI Matching → Work Proof Verification → Risk Prediction → Automated Invoicing

Technical Approach & Architecture

The platform features a modern, production-ready decoupled architecture built to handle enterprise-level scale and strict security guidelines:

- **Frontend Layer:** Built using an AI-architected React UI featuring tailored role-based dashboards for both agents and contractors. Built leveraging modern rapid UI generation engines (Lovable) for high responsive layout accuracy.
- **Backend & Database Layer:** Powered by **FastAPI (Python)** and an AI-orchestrated **Supabase** engine. It utilizes a PostgreSQL core enforcing *Row Level Security (RLS)* for absolute role-based isolation and real-time data streaming via granular APIs.

- **Document Processing Pipeline:** Employs reliable PDF generation libraries (ReportLab/PDFKit) to auto-convert field logs and progress entries directly into standardized, legally compliance invoices.
- **Security Layer:** Implements industry-grade TLS 1.3 encrypted communication protocols, hardened user authentication, and multi-factor safety controls.

Project Impact & Business Viability

Feasibility Metrics

- **Modular Design:** Decoupled layers allow independent updates with zero operational downtime.
- **10x Scalability:** AI dispatch matching eliminates the administrative overhead scaling ceilings.
- **Data-Driven Reliability:** The continuous machine learning loop enhances tracking and risk prediction accuracy over time.

Strategic Value Benefits

- **Economic ROI:** Drives down operational overhead, completely eliminates billing errors, and protects project margins.
- **Social / Gig Economy:** Establishes a pure merit-based allocation system with complete financial transparency.
- **Zero-Trust Security:** Ensures 100% reliable tracking via geo-fenced boundaries.

Budgetary Overview & Milestones

Current project status is **30% Complete**. Core items completed include UI/UX wireframes, complete database relational schema design, user authentication, and basic job posting routes.

Category	Description	Estimated Cost Structure
Development	UI customization & workflow logic tweaks	₹3,000 - ₹5,000
Development	Testing & detailed bug fixing iterations	₹2,000 - ₹3,000
Development	Documentation & interactive demo preparation	₹1,000 - ₹2,000
Lovable AI / Infrastructure	Rapid UI generation & prototyping (Free tier)	₹0
Supabase Backend	Database, Secure Authentication, and RLS (Free tier)	₹0
Supabase Storage	Secure image/video field progress storage	₹0 - ₹500
Development Hosting	Preview staging and hosting	₹0
Infrastructure	Custom professional domain acquisition	₹800 - ₹1,500

Category	Description	Estimated Cost Structure
TOTAL	**Overall project development budget**	**Locally Scaled: ₹6,500 - ₹12,500**

Project Team Profile

All members are affiliated with **Sri Ramakrishna College of Arts & Science, Coimbatore**:

Name	Core Project Role	Gender
SUNDARJI S	TEAM LEADER	Male
HARSHINI C	DEVELOPER	Female
GOPI KRISHNA M	DEVELOPER	Male
DINESH KARTHICK R	RESEARCHER	Male
GOKULAVANI V	RESEARCHER	Female
KAVIYAPRIYA R	PRESENTER	Female